EXPERIMENT 1.3

Mapped Course Outcomes-CO2

CO2: Understand the working of routers, switches and hubs

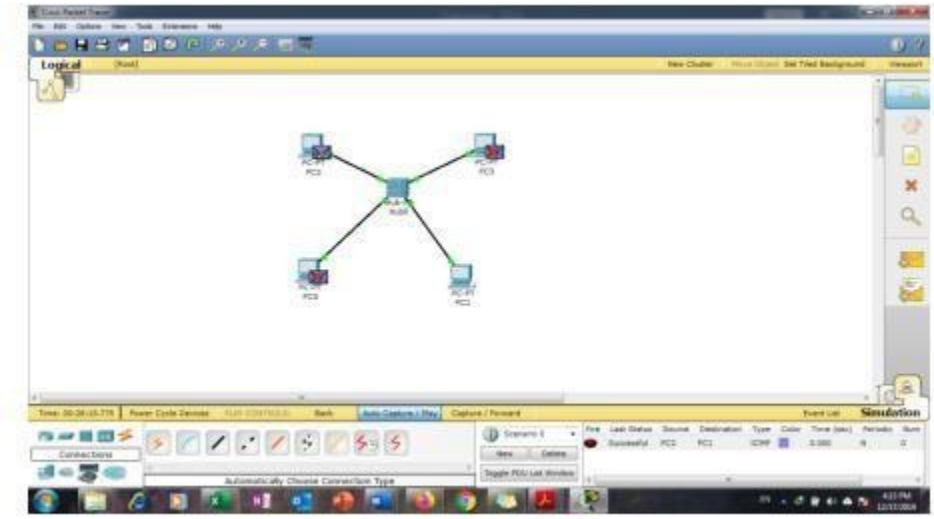
Aim-

Configure and understand working of network devices hub,switch and router..

Procedure-

1. Attach required devices (Hub/Switch/Router) in the packet tracer software.
2. Assign IP address to devices.
3. Select source and destination and drop packet from source to destination.
4. Go to Simulation mode and click capture/Play.
5. Simulation will start and packet will only be accepted by destination

SAMPLE OUTPUT:



Procedure: Following should be done to understand this practical.

1. Repeater:Functioning at Physical Layer.Arepeater is an electronic device

that receives a signal and retransmits it at a higher level and/or higher power, or onto

the other side of an obstruction, so that the signal can cover longer distances. Repeater have two ports ,so cannot be use to connect for more than two devices

2. Hub: An Ethernet hub, active hub, network hub, repeater hub, hub or concentrator is a device for connecting multiple twisted pair or fiber optic Ethernet devices together and making them act as a single network segment. Hubs work at the physical layer (layer 1) of the OSI model. The device is a form of multiport repeater. Repeater hubs also participate in collision detection, forwarding a jam signal to all ports if it detects a collision.

3. Switch:Anetwork switch or switching hub is a computer networking device that connects network segments.The term commonly refers to a network bridge that processes and routes data at the data link layer (layer 2) of the OSI model. Switches that additionally process data at the network layer (layer 3 and above) are often referred to as Layer 3 switches or multilayer switches.

4. Bridge: A network bridge connects multiple network segments at the data link layer (Layer 2) of the OSI model. In Ethernet networks, the term bridge formally means a device that behaves according to the IEEE 802.1D standard. A bridge and switch are very much alike; a switch being a bridge with numerous ports. Switch or Layer 2 switch is often used interchangeably with bridge .Bridges can analyze incoming data packets to determine if the bridge is able to send the given packet to another segment of the network.

5. Router: A router is an electronic device that interconnects two or more computer networks, and selectively interchanges packets of data between them. Each data packet contains address information that a router can use to determine if the source and destination are on the same network, or if the data packet must be transferred from one network to another. Where multiple routers are used in a large

collection of interconnected networks, the routers exchange information about

target system addresses, so that each router can build up a table showing the

preferred

paths between any two systems on the interconnected networks.

6. Gate Way: In a communications network, a network node equipped for interfacing withanother network that uses different protocols.

- A gateway may contain devices such as protocol translators, impedance

matching devices, rate converters, fault isolators, or signal translators as

necessary to provide system interoperability. It also requires the establishment

of mutually acceptable administrative procedures between both networks.

- A protocol translation/mapping gateway interconnects networks

with different network protocol technologies by performing the required

protocol conversions

Further Reading

7. https://geekflare.com/understanding-ip-address/
8. https://docs.oracle.com/cd/E19504-01/802-5753/planning3-18471/index.html
9. https://www.ibm.com/cloud/learn/networking-a-complete-guide

Research Articles:

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6. **h ttp s://www.tand fonl ine.com/d oi/ full /10.1080/ 21642850.2018.1521283**